

Lecture 12 – Ideal gas, work and heat

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In this lecture we consider a **thermodynamic system** (e.g., an **ideal gas**) of a very large but fixed number N of (non-interacting, identical) “particles” which are **atoms** or **molecules** each with mass m , that are in **random motion**. It is impossible to describe such a system by specifying the positions and velocities of the “particles” of the system due to the persistent changes in random manner. Instead, we describe the system by specifying the **pressure** p , **temperature** T , and **volume** V of the system. We also consider how these variables may change by doing **work** on and/or supplying **heat** to the system.

A few slides in this chapter come from Chapter-18 of Young’s textbook.
This PPT file will well define the scope of your tests.

In this chapter **pressure** is denoted by p .
Please do not mix up with momentum

1



Random Motion

Impossible to specify the position and speed of the particles because the values keep changing and they are in random directions.

2

Number of Moles



In a sample with two grams (i.e., 2 g) of hydrogen gas, there are **Avogadro's number** of hydrogen molecules:

Avogadro's number, $N_A \approx 6.022 \times 10^{23}$ particles/mole.

This is also the number of carbon-12 atoms in a sample of 12 g of carbon-12.

This is also the number of water molecules in a sample of 18 g of water.

This number of particles is referred to as 1 mole of the substance.

The number of particles (atoms/molecules) in **1 mole** of a substance is the same for any substance.

Remark.

How big is 6.022×10^{23} in 1 mole? *Very big!!*

The age of our universe is approximately 13.8×10^9 years!!

3

Mass and molar mass

For a sample of a substance where there are N identical particles, each of mass m , the total mass (M) of the sample is $M = N \times m$ (kg).

So, **one mole** of the substance has molar mass $M_{\text{molar}} = N_A m$ (kg/mole).

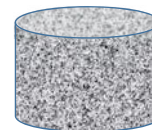
This is called the **molar mass** of the substance where you can calculate with the help of periodic table.

E.g., the **molar mass** of carbon-12 (commonest natural carbon isotope) is approximately 0.012 kg/mole, and the **molar mass** of water is approximately 0.018 kg/mole.

For a sample of a substance with mass M , the amount, i.e., **number of moles** (n) of substance in the sample is

$$n = \frac{M}{M_{\text{molar}}} = \frac{N}{N_A} \text{ (mole).}$$

Diagram labels: mass in sample points to M ; molar mass points to M_{molar} ; number of particles in sample points to N ; 6.022×10^{23} points to N_A .



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1 mole of carbon-12 has 12 g or 0.012 kg.

Water: H_2O
 $1 \times 2 + 16 = 18 \text{ u}$
 1 mole of water has 18 g or 0.018 kg.

Slight deviation because the mass of proton deviates slightly from the mass of neutron.

Key:

Symbol	Al	13	Atomic Number
Atomic Name	Aluminum	26.982	Atomic Weight

Courtesy: <https://www.fishersci.com/us/en/periodic-table.html>

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Constant, random motion, and root-mean-square speed

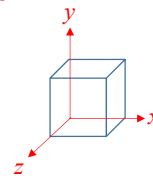
Consider a sample of an ideal gas of a total mass M , where there are N identical molecules each of mass m , where $m = \frac{M}{N}$, in constant and **random motion**.

The total kinetic energy associated with the **random translational motion** of N molecules is given by the summation where v is 3-directional:

$$KE_{\text{translation}} = \sum \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \sum v^2 = \frac{1}{2} \left(\frac{M}{N} \right) \sum v^2 = \frac{1}{2} M \left(\frac{\sum v^2}{N} \right)$$

where $v^2 = v_x^2 + v_y^2 + v_z^2$

by Pythagorean theorem.



Because x, y, z are 90° (independent) to each others, $\sum v^2 = \left(\sum v_x^2 + \sum v_y^2 + \sum v_z^2 \right)$

$$\Rightarrow \frac{1}{N} \sum v^2 = \frac{1}{N} \sum v_x^2 + \frac{1}{N} \sum v_y^2 + \frac{1}{N} \sum v_z^2.$$

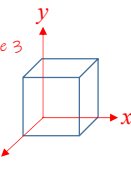
6

Constant, random motion, and root-mean-square speed (cont.)

By **random translational motion** and considering the **average** values in all **independent** directions, and, for **big** N ,

$$\frac{1}{N} \sum v_x^2 = \frac{1}{N} \sum v_y^2 = \frac{1}{N} \sum v_z^2.$$

The average of **large** sample for these 3 components are equal to each other.



$$v^2 = v_x^2 + v_y^2 + v_z^2$$

So, $\frac{1}{N} \sum v_x^2 = \frac{1}{N} \sum v_y^2 = \frac{1}{N} \sum v_z^2 = \frac{1}{N} \sum \frac{v^2}{3} = \frac{1}{3} \left(\frac{\sum v^2}{N} \right) = \frac{1}{3} v_{\text{rms}}^2$ for big N ,

where the **root-mean-square speed** $v_{\text{rms}} = \sqrt{\frac{\sum v^2}{N}}.$

Mean of {4, 7, 8} is $\frac{4+7+8}{3} = 6.33$

Root mean square of {4, 7, 8} is $\sqrt{\frac{4^2+7^2+8^2}{3}} = 6.56$

Therefore, $KE_{\text{translation}} = \frac{1}{2} M \left(\frac{\sum v^2}{N} \right) = \frac{1}{2} M \left(\sqrt{\frac{\sum v^2}{N}} \right)^2 = \frac{1}{2} M v_{\text{rms}}^2$

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State variables, equation of state, and the ideal gas equation

What are your states now?

Amount of substance denoted as n (number of moles), **pressure** p , **temperature** T , and **volume** V are examples of **state variables**. They describe the conditions in which a particular substance exists - they describe the “**state**” of the substance.

The volume V of a substance is usually determined by its pressure p , temperature T , and amount of substance n (number of moles). In general, we can't change one of these variables without causing a change in another - we can express the relationship among n , p , T , and V as an equation, using graphs or numerical tables - the **equation of state**.

For example, for a sample of n mole of an **ideal gas**, we have the **ideal gas equation**,
(also called **universal gas law**)

$$pV = nRT$$

where **gas constant** $R \approx 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

Equivalently, $pV = \frac{N}{N_A} RT = NkT$, where **Boltzmann constant**

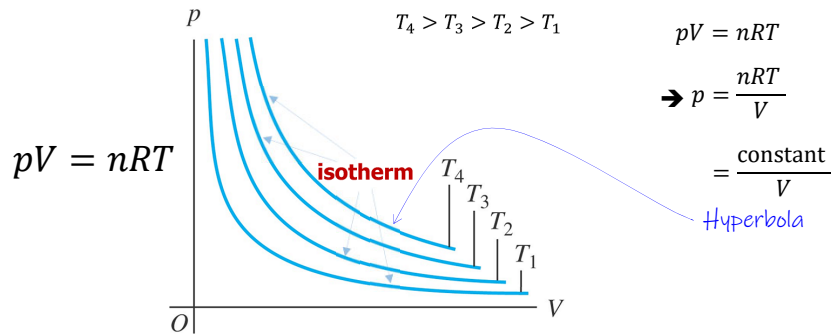
$$k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} \approx 1.381 \times 10^{-23} \text{ J/K}.$$

Number of particles in the sample

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State variables, equation of state, and the ideal gas equation (cont.)

Remark – the **pV -diagram**.



Each curve (**isotherm**) represents pressure p as a function of volume V for an (n mole) **ideal gas** at a **single** temperature T – for each curve (**isotherm**) pV is a constant that is directly proportional to T .

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State variables, equation of state, and the ideal gas equation (cont.)

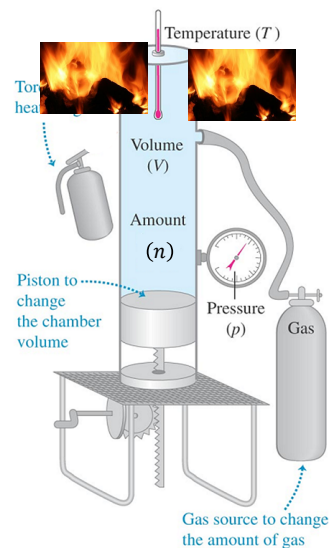
Remark – the **ideal gas equation**, $pV = nRT$.

In general, we can have 4 simultaneous variables p, V, n, T in the ideal gas equation $pV = nRT$.

Keeping p and T constant, $V \propto n$.

Keeping n and T constant, $V \propto \frac{1}{p}$.

Keeping n and V constant, $p \propto T$.



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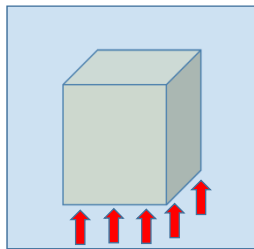
Problem. The mass of a hot-air balloon and its cargo (not including the air inside) is 200 kg. The air outside is at 10.0°C and 101 kPa. The volume of the balloon is 400 m³. To what temperature must the air in the balloon be heated before the cargo will lift off? (Air density at 10.0°C is 1.25 kg/m³)

Answer:

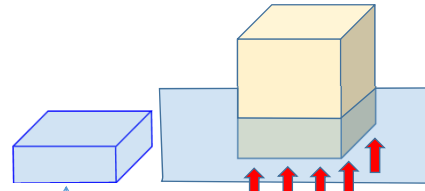
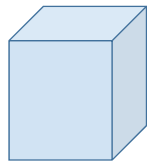
Recall the **Archimedes' principle** for floatation:

“Any body completely or partially submerged in a fluid (gas or liquid) at rest is acted upon by an upward force (called upthrust or buoyant force) of magnitude equal to the weight of the fluid displaced by the body.”

The volume of displaced fluid is equivalent to the volume of an object fully immersed in a fluid, or, to that fraction of the volume below the fluid surface for an object partially submerged in a liquid.



The magnitude of the upthrust is equal to the magnitude of the weight of the fluid in this volume.



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Problem. The mass of a hot-air balloon and its cargo (not including the air inside) is 200 kg. The air outside is at 10.0°C and 101 kPa. The volume of the balloon is 400 m³. To what temperature must the air in the balloon be heated before the cargo will lift off? (Air density at 10.0°C is 1.25 kg/m³)

Answer:

Initially the balloon has displaced V volume of the air outside. By **Archimedes' principle**, the upthrust provided by the air outside is $(\rho_{\text{outside}} \times V)g$.

But initially the density of the air inside the balloon is same as the density of the air outside. As the upthrust is now balanced by total downward weight of air inside, the cargo (with basket) remains seated on the ground (Here we assume that the canvas of the balloon is very light. In the reality the balloon will have to be slightly heated to sustain the weight of the canvas so that the balloon will remain stationary).

As the air inside the balloon is heated, the mass of the air inside the balloon decreases due to the increased kinetic energy of the air molecules making it escape. Let the density of the air in the heated

balloon be $\rho_{\text{inside}}^{\text{heated}}$. From the **ideal gas equation** $PV = nRT$, the molar density (mol/m³) of the air is $\frac{n}{V} = \frac{P}{R[T]}$.

But density = molar density $\times$ molar mass. Thus: density $\propto \frac{1}{T}$. So $\rho_{\text{inside}}^{\text{heated}} \times T_{\text{inside}}^{\text{heated}} = \rho_{\text{outside}} \times T_{\text{outside}} \Rightarrow \rho_{\text{inside}}^{\text{heated}} = \rho_{\text{outside}} \times \frac{T_{\text{outside}}}{T_{\text{inside}}^{\text{heated}}}$.

Net force on balloon without Cargo : upthrust - $(\rho_{\text{inside}}^{\text{heated}} \times V) \times g = (\rho_{\text{outside}} \times V) \times g - (\rho_{\text{inside}}^{\text{heated}} \times V) \times g$

$$= (\rho_{\text{outside}} \times V) \times g - \left(\rho_{\text{outside}} \times \frac{T_{\text{outside}}}{T_{\text{inside}}^{\text{heated}}} \times V \right) \times g > 200 \text{ kg} \times g$$

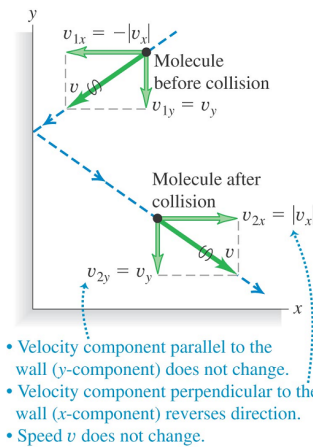
$$\Rightarrow (1.25 \text{ kg/m}^3)(400 \text{ m}^3) \left(1 - \frac{10 + 273.15}{T_{\text{inside}}^{\text{heated}}} \right) > 200 \text{ kg} \Rightarrow T_{\text{inside}}^{\text{heated}} > 472.42 \text{ K } (199.27^\circ\text{C})$$

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Kinetic Theory on Ideal Gases

Pascal

Pressure and work done



Assumption: The walls of container are perfectly rigid, infinitely massive and do not move.

Pressure p (1 Pa = 1 N/m²) refers both to the force per unit area exerted by the gas on the walls of the container, and to the force per unit area exerted by the walls of the container back on the gas. (which one is action and which one is reaction is not important from Newton's 3rd Law.)

Each molecule of the gas collides **occasionally** with a wall of the container. During collisions the molecules exert forces on the walls of the container - this is the origin of the **pressure** that the gas exerts.

These **collisions** are **perfectly elastic**.

Change in momentum along x-axis for 1 particle: $\leftarrow +$

$$dP_x = mv_f - mv_i = (-mv_x) - mv_x = -2mv_x$$

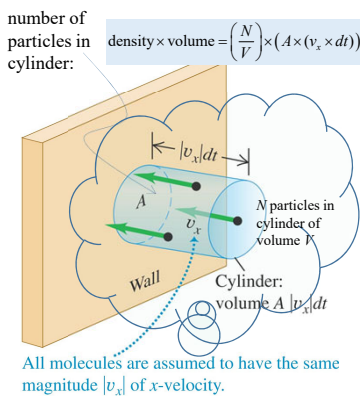
Change in momentum along y-axis for 1 particle: $\uparrow +$

$$dP_y = mv_f - mv_i = (-mv_y) - (-mv_y) = 0$$

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Kinetic Theory on Ideal Gases

Pressure and work done (cont.)



On the average, half of the particles in the cylinder are moving toward the wall, and half are moving from it.

The number of collisions on area A is $\frac{\text{no. of particles in cylinder}}{2} = \frac{\left(\frac{N}{V}\right) \times (A \times (v_x \times dt))}{2}$.

Impulse exerted on all particles by the small circular wall A = changes in linear momentum

$$\rightarrow F \times dt = (-2mv_x) \times \frac{\left(\frac{N}{V}\right) \times (A \times (v_x \times dt))}{2}$$

By Newton's 3rd law, there will be impulse of equal magnitude exerted on the wall by all the particles but in **opposite** direction. So the pressure on the wall is

$$v_x^2 = \frac{v_{\text{rms}}^2}{3}$$

$$P = -\frac{F}{A} = -\frac{(-2mv_x) \times \left(\frac{N}{V}\right) \times (A \times (v_x \times dt))}{A dt} = \frac{Nmv_x^2}{V} = \frac{Mv_x^2}{V} = \left(\frac{M}{V}\right) \frac{v_{\text{rms}}^2}{3} = \frac{1}{3} \left(\frac{M}{V}\right) v_{\text{rms}}^2$$

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Kinetic Theory on Ideal Gases

Pressure and work done (cont.)

Remark.

$$\text{Pressure } p = \frac{1}{3} \left(\frac{M}{V} \right) v_{rms}^2 = \frac{2}{3V} \left(\frac{1}{2} M v_{rms}^2 \right) = \frac{2}{3V} KE_{\text{translation}}.$$

$$\Rightarrow nRT = pV = \frac{2}{3} KE_{\text{translation}} \Rightarrow KE_{\text{translation}} = \frac{3}{2} nRT.$$

For only one particle of mass m

$$\frac{1}{2} m v_{rms}^2 = \frac{3}{2} \frac{nRT}{N} = \frac{3}{2} \left(\frac{R}{N_A} \right) T = \frac{3}{2} kT.$$

$$\Rightarrow v_{rms} = \sqrt{\frac{3kT}{m}}.$$

number of particles in cylinder N

number of moles n

$$N = n N_A$$

$$\rightarrow \frac{n}{N} = \frac{1}{N_A}$$

6.022×10^{23}

Boltzmann constant

$$k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} \approx 1.381 \times 10^{-23} \text{ J/K}$$

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Example. An oxygen molecule is moving in the ionosphere (the upper most part of the earth's atmosphere) where the Kelvin temperature is 3 times of the temperature near the earth's surface. Determine the ratio of the translational rms speed in the ionosphere to that near the earth's surface for the oxygen molecule.

Answer:

Based on $v_{rms} = \sqrt{\frac{3kT}{m}},$

Boltzmann constant

$$k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} \approx 1.381 \times 10^{-23} \text{ J/K}$$

$$\frac{(v_{rms})_{\text{ionosphere}}}{(v_{rms})_{\text{earth's surface}}} = \frac{\sqrt{\frac{3kT_{\text{ionosphere}}}{m}}}{\sqrt{\frac{3kT_{\text{earth's surface}}}{m}}} = \sqrt{\frac{T_{\text{ionosphere}}}{T_{\text{earth's surface}}}} = \sqrt{3} = 1.73$$

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Summary of the Formulation in Kinetic Theory for Ideal Gas

Avogadro's number, $N_A \approx 6.022 \times 10^{23}$ particles/mole.

→ 1 mole of any substance kept in a container has 6.022×10^{23} particles in the container



Molar mass is the total mass (in kg) of the substance when it has 6.022×10^{23} particles.

The translational kinetic energy of n moles of substance is only dependent on its temperature.

$$KE_{\text{translation}} = \frac{3}{2} nRT.$$

Kinetic energy for only one particle of mass m

$$\frac{1}{2} m v_{\text{rms}}^2 = \frac{3}{2} \frac{nRT}{N} = \frac{3}{2} \left(\frac{R}{N_A} \right) T = \frac{3}{2} kT.$$

Annotations: "number of moles in container" points to n ; "number of particles in container" points to N .

number of particles in cylinder $\rightarrow N = n N_A$
 number of moles $\rightarrow n$
 $\rightarrow \frac{n}{N} = \frac{1}{N_A}$
 6.022×10^{23}

Boltzmann constant

$$k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} \approx 1.381 \times 10^{-23} \text{ J/K}$$

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Question. A cylinder contains a mixture of helium and argon gas in equilibrium at 150°C .

Boltzmann constant $k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} \approx 1.381 \times 10^{-23} \text{ J/K}$

(i) What is the average kinetic energy for each type of gas molecule?

Answer:

Both helium and argon have the same average kinetic energy because they have the same temperature.

From $\frac{1}{2} m v_{\text{rms}}^2 = \frac{3}{2} kT$. *Boltzmann constant*

$$\overline{KE} = \frac{3}{2} (1.38 \times 10^{-23} \text{ J/K}) (150 + 273.15 \text{ K}) = 8.76 \times 10^{-21} \text{ J}$$

(But their speed will be different because their mass is different!!)

(ii) What is the root-mean-square speed of each type of molecule?

Answer:

$$\overline{KE} = \frac{1}{2} m v_{\text{rms}}^2 = 8.76 \times 10^{-21} \text{ J} \rightarrow v_{\text{rms}} = \sqrt{\frac{2 \times (8.76 \times 10^{-21}) \text{ J}}{m}} = \sqrt{\frac{1.75 \times 10^{-20} \text{ J}}{m}} \dots (1)$$

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... cont

2	He	Helium	4.003
10	Ne	Neon	20.18
18	Ar	Argon	39.95
36	Kr	Krypton	83.80

For helium, $m = \frac{4.003 \text{ g/mol}}{6.02 \times 10^{23} \text{ molecules/mol}} = 6.64 \times 10^{-24} \text{ g/molecule}$

$$\rightarrow m_{\text{helium}} = 6.64 \times 10^{-27} \text{ kg/molecule}$$

For argon, $m = \frac{39.95 \text{ g/mol}}{6.02 \times 10^{23} \text{ molecules/mol}} = 6.63 \times 10^{-23} \text{ g/molecule}$

$$\rightarrow m_{\text{argon}} = 6.63 \times 10^{-26} \text{ kg/molecule}$$

Substituting in (1),

For helium, $v_{\text{rms}} = \sqrt{\frac{1.75 \times 10^{-20} \text{ J}}{6.64 \times 10^{-27} \text{ kg}}} = 1624 \text{ m/s}$

$$v_{\text{rms}} = \sqrt{\frac{1.75 \times 10^{-20} \text{ J}}{m}} \dots (1)$$

Kinetic Theory
for Ideal Gas !!

For argon, $v_{\text{rms}} = \sqrt{\frac{1.75 \times 10^{-20} \text{ J}}{6.63 \times 10^{-26} \text{ kg}}} = 514 \text{ m/s}$

$$\text{or } \frac{1}{2} m v_{\text{rms}}^2 = \frac{3}{2} kT \rightarrow v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

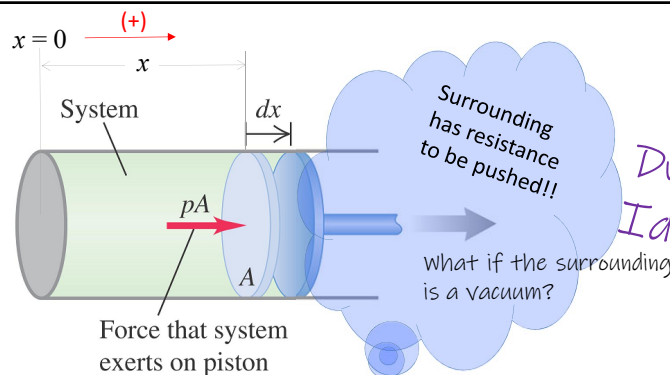
For helium $v_{\text{rms}} = \sqrt{\frac{3 \times (1.38 \times 10^{-23}) \times (150 + 273.15) \text{ J}}{6.64 \times 10^{-27} \text{ kg}}} = 1624 \text{ m/s}$

For argon $v_{\text{rms}} = \sqrt{\frac{3 \times (1.38 \times 10^{-23}) \times (150 + 273.15) \text{ J}}{6.63 \times 10^{-26} \text{ kg}}} = 514 \text{ m/s}$

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Pressure and work
done (cont.)

(Treat going right as positive)

Dynamics for
Ideal Gas !!

Now consider a sample of an ideal gas enclosed in a container with a movable piston. When the gas expands, it pushes outward on its boundary surfaces as they move outward – an **expanding gas** always does **positive work** (on its

surroundings): $dW = Fdx = \left(\frac{F}{A}\right) A dx = p A dx = p dV$.

Question.

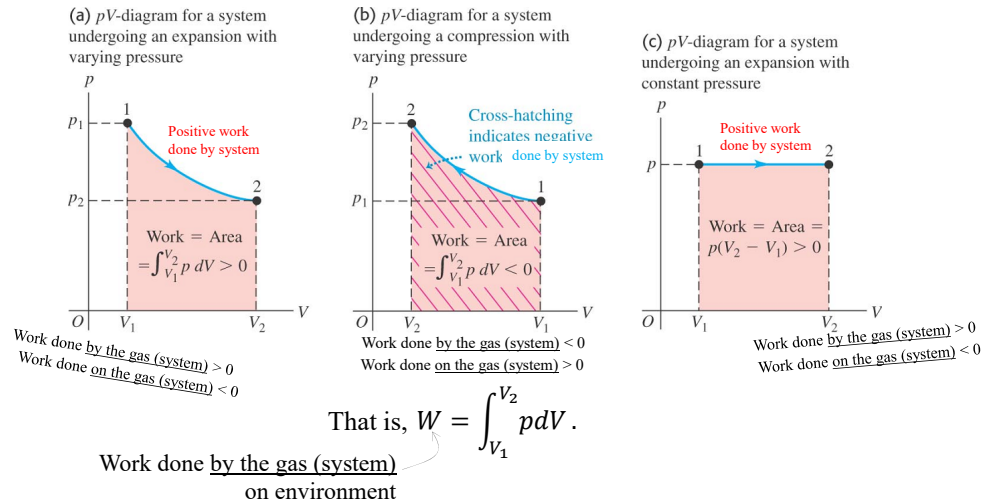
The molecules in an expanding gas lose $KE_{\text{translation}}$ and the gas cools down after the expansion. How?

$$\frac{1}{2} m v_{\text{rms}}^2 = \frac{3}{2} kT$$

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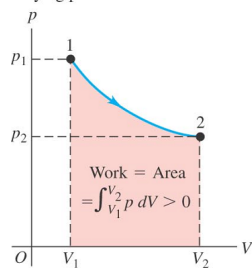
Pressure and work done (cont.)

In a finite change of volume from V_1 to V_2 , the work done equals the area under the curve on a pV -diagram:



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(a) pV -diagram for a system undergoing an expansion with varying pressure

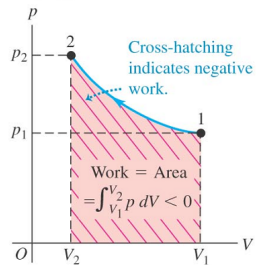


Work done by the gas (system) > 0
 Work done on the gas (system) < 0

Expansion

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(b) pV -diagram for a system undergoing a compression with varying pressure

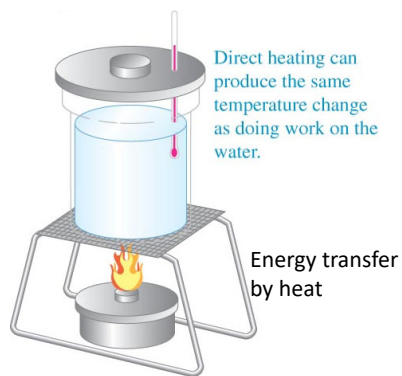


Work done by the gas (system) < 0
 Work done on the gas (system) > 0

Compression

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Temperature and heat

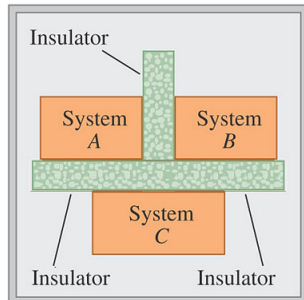


Energy transfer from one substance to another can also take place solely because of a **temperature difference**. This is called **energy transfer** by **heat flow** – this is also called **heat transfer** and it causes temperature changes. The energy transferred in this way is called **heat**.

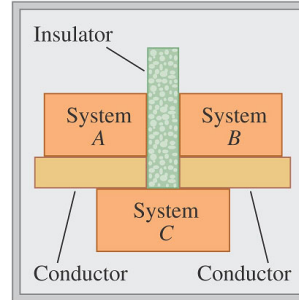
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Temperature and heat (cont.)

Temperature is the quantity that determines if **heat transfer** is possible.



perfect
insulating box

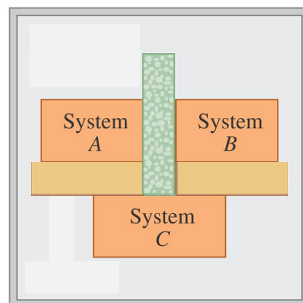


Consider three systems A, B, and C, which are initially not in **thermal equilibrium**. Due to the (ideal) insulating box, they cannot interact with anything except with each other.

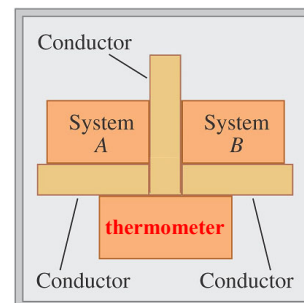
Now, let C interact with both A and B, while A and B are separated by an (ideal) insulating wall – after some time A and B are each in **thermal equilibrium** with C. Are A and B in **thermal equilibrium** with each other?

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Temperature and heat (cont.)



perfect
insulating box



If C is in **thermal equilibrium** with both A and B, then A and B are also in **thermal equilibrium** with each other

– **zeroth law of thermodynamics**

Two systems are in **thermal equilibrium** if and only if they have the same **temperature**.

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Temperature and heat (cont.)

Remark – Kelvin or absolute temperature.

T in the **ideal gas equation**, $pV = nRT$, is the **Kelvin (K)** or **absolute temperature**:

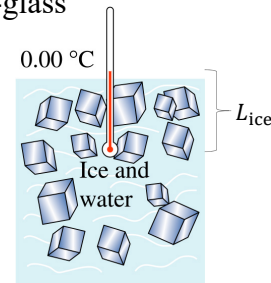
$$T = T_{\text{Celsius}} + 273.15.$$

Here, T_{Celsius} is the **Celsius temperature**. Clearly, $\Delta T = \Delta T_{\text{Celsius}}$.

Many measurable properties of matter depend on temperature – for example the length a mercury thread in a mercury-in-glass thermometer. The **Celsius temperature scale**:

$$T_{\text{Celsius}}(^{\circ}\text{C}) = \frac{L_T - L_{\text{ice}}}{L_{\text{steam}} - L_{\text{ice}}} \times 100^{\circ}\text{C}$$

Linear Calibration
(L refers to the length of the mercury thread.)



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Specific heat and molar heat capacity

The quantity of heat Q required to increase the temperature of a mass M of a certain substance from T_1 to T_2 is $Q = Mc(T_2 - T_1)$,

where c is the **specific heat** of the substance.
 c has SI unit, $\text{J kg}^{-1}\text{K}^{-1}$.

$$Q = Mc(T_2 - T_1) = \left(\frac{\overset{\text{number of moles } n}{M}}{\underset{\text{Capital letter } C}{M_{\text{molar}}}} \right) M_{\text{molar}} c (T_2 - T_1) = nC(T_2 - T_1),$$

where C is the **molar heat capacity** of the substance. C has SI unit $\text{J mol}^{-1}\text{K}^{-1}$.

$$\begin{aligned} M_{\text{molar}} C &= M_{\text{molar}} \times \frac{\text{heat}}{\text{mass} \cdot \Delta \text{temperature}} \\ &= \frac{\text{mass} \cdot n}{M_{\text{molar}}} \cdot \Delta \text{temperature} \\ &= C \quad (\text{molar heat capacity}) \end{aligned}$$

For an infinitesimal change in temperature dT , $\frac{dQ}{dt} = \frac{Mc}{dt}dT$, or $\frac{dQ}{dt} = \frac{nC}{dt}dT$.

$$\text{So, } c = \frac{1}{M} \left(\frac{dQ}{dT} \right), \text{ and } C = \frac{1}{n} \left(\frac{dQ}{dT} \right).$$

If dT takes place over an infinitesimal **time interval** dt , then

$$\frac{dQ}{dt} = Mc \left(\frac{dT}{dt} \right), \quad \text{or} \quad \frac{dQ}{dt} = nC \left(\frac{dT}{dt} \right).$$

$\frac{dT}{dt}$: Temperature gradient w.r.t. time

Why are c and C also called thermal inertia?

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Specific heat and molar heat capacity (cont.)

Table 17.3 Approximate Specific Heats and Molar Heat Capacities
(Constant Pressure)

Substance	Specific Heat, c_p (J/kg · K)	Molar Mass, M_{molar} (kg/mol)	Molar Heat Capacity, C_p (J/mol · K)
Aluminum	910	0.0270	24.6
Beryllium	1970	0.00901	17.7
Copper	390	0.0635	24.8
Ethanol	2428	0.0461	111.9
Ethylene glycol	2386	0.0620	148.0
Ice (near 0°C)	2100	0.0180	37.8
Iron	470	0.0559	26.3
Lead	130	0.207	26.9
Marble (CaCO ₃)	879	0.100	87.9
Mercury	138	0.201	27.7
Salt (NaCl)	879	0.0585	51.4
Silver	234	0.108	25.3
Water (liquid)	4190	0.0180	75.4

Specific Heat
Capacity at
Constant PressureMolar Heat
Capacity at
Constant PressureConstant pressure → Do not constrain the
expansion or contraction due to the heat

$$\frac{dQ}{dt} = Mc \left(\frac{dT}{dt} \right) = nC \left(\frac{dT}{dt} \right)$$

$$\rightarrow C = \left(\frac{M}{n} \right) c$$

$$= \left(\frac{\text{Total Mass equivalent to 1 mole}}{6.022 \times 10^{23}} \right) c$$

$$\rightarrow C = M_{\text{molar}} c$$

Molar Heat Capacity = Molar Mass x Specific Heat Capacity

$$\text{NaCl: } 51.4 = \left(\frac{22.99 + 35.453}{1000} \text{ kg} \right) \times 879$$

$$\text{H}_2\text{O: } 1 \times 2 + 16 = 18 \text{ g} = 0.018 \text{ kg}$$

$$75.4 = 0.018 \text{ kg} \times 4190$$

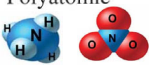
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H 1 Hydrogen 1.008 Li 3 Lithium 6.941 Na 11 Sodium 22.99 Be 4 Beryllium 9.012 Mg 12 Magnesium 24.31																B 5 Boron 10.81 C 6 Carbon 12.01 N 7 Nitrogen 14.01 O 8 Oxygen 16.00 F 9 Fluorine 19.00 Ne 10 Neon 20.18 Al 13 Aluminum 26.98 Si 14 Silicon 28.09 P 15 Phosphorus 30.97 S 16 Sulfur 32.07 Cl 17 Chlorine 35.45 Ar 18 Argon 39.95															
K 19 Potassium 39.10 Ca 20 Calcium 40.08 Sc 21 Scandium 44.96 Ti 22 Titanium 47.87 V 23 Vanadium 50.94 Cr 24 Chromium 52.00 Mn 25 Manganese 54.94 Fe 26 Iron 55.85 Co 27 Cobalt 58.93 Ni 28 Nickel 58.69 Cu 29 Copper 63.55 Zn 30 Zinc 65.38 Ga 31 Gallium 69.72 Ge 32 Germanium 72.63 As 33 Arsenic 74.92 Se 34 Selenium 78.97 Br 35 Bromine 79.90 Kr 36 Krypton 83.80																															
Rb 37 Rubidium 85.47 Sr 38 Strontium 87.62 Y 39 Yttrium 88.91 Zr 40 Zirconium 91.22 Nb 41 Niobium 92.91 Mo 42 Molybdenum 95.95 Tc 43 Technetium 98.00 Ru 44 Ruthenium 101.1 Rh 45 Rhodium 102.9 Pd 46 Palladium 106.4 Ag 47 Silver 107.9 Cd 48 Cadmium 112.4 In 49 Indium 114.8 Sn 50 Tin 118.7 Sb 51 Antimony 121.8 Te 52 Tellurium 127.6 I 53 Iodine 126.9 Xe 54 Xenon 131.3																															
<div><div><div><div>Cs</div><div>55</div></div><div>Cesium</div><div>132.9</div></div><div><div><div>Ba</div><div>56</div></div><div>Barium</div><div>137.3</div></div><div><div><div colspan="2">Lanthanides</div></div></div><div><div><div>Hf</div><div>72</div></div><div>Hafnium</div><div>178.5</div></div><div><div><div>Ta</div><div>73</div></div><div>Tantalum</div><div>180.9</div></div><div><div><div>W</div><div>74</div></div><div>Tungsten</div><div>183.8</div></div><div><div><div>Re</div><div>75</div></div><div>Rhenium</div><div>186.2</div></div><div><div><div>Os</div><div>76</div></div><div>Osmium</div><div>190.2</div></div><div><div><div>Ir</div><div>77</div></div><div>Iridium</div><div>192.2</div></div><div><div><div>Pt</div><div>78</div></div><div>Platinum</div><div>195.1</div></div><div><div><div>Au</div><div>79</div></div><div>Gold</div><div>197.0</div></div><div><div><div>Hg</div><div>80</div></div><div>Mercury</div><div>200.6</div></div><div><div><div>Tl</div><div>81</div></div><div>Thallium</div><div>204.4</div></div><div><div><div>Pb</div><div>82</div></div><div>Lead</div><div>207.2</div></div><div><div><div>Bi</div><div>83</div></div><div>Bismuth</div><div>209.0</div></div><div><div><div>Po</div><div>84</div></div><div>Polonium</div><div>(209)</div></div><div><div><div>At</div><div>85</div></div><div>Astatine</div><div>(210)</div></div><div><div><div>Rn</div><div>86</div></div><div>Radon</div><div>(222)</div></div></div>																															
<div><div><div><div>Fr</div><div>87</div></div><div>Francium</div><div>(223)</div></div><div><div><div>Ra</div><div>88</div></div><div>Radium</div><div>(226)</div></div><div><div><div colspan="2">Actinides</div></div></div><div><div><div>Rf</div><div>104</div></div><div>Rutherfordium</div><div>(261)</div></div><div><div><div>Db</div><div>105</div></div><div>Dubnium</div><div>(268)</div></div><div><div><div>Sg</div><div>106</div></div><div>Seaborgium</div><div>(269)</div></div><div><div><div>Bh</div><div>107</div></div><div>Bohrium</div><div>(270)</div></div><div><div><div>Hs</div><div>108</div></div><div>Hassium</div><div>(269)</div></div><div><div><div>Mt</div><div>109</div></div><div>Meitnerium</div><div>(278)</div></div><div><div><div>Ds</div><div>110</div></div><div>Darmstadtium</div><div>(281)</div></div><div><div><div>Rg</div><div>111</div></div><div>Roentgenium</div><div>(281)</div></div><div><div><div>Cn</div><div>112</div></div><div>Copernicium</div><div>(285)</div></div><div><div><div>Nh</div><div>113</div></div><div>Nihonium</div><div>(286)</div></div><div><div><div>Fl</div><div>114</div></div><div>Flerovium</div><div>(289)</div></div><div><div><div>Mc</div><div>115</div></div><div>Moscovium</div><div>(289)</div></div><div><div><div>Lv</div><div>116</div></div><div>Livermorium</div><div>(293)</div></div><div><div><div>Ts</div><div>117</div></div><div>Tennessine</div><div>(294)</div></div><div><div><div>Og</div><div>118</div></div><div>Oganesson</div><div>(294)</div></div></div>																															
Key Al 13 Atomic Number Al Aluminum 26.982 Atomic Weight La 57 Lanthanum 138.9 Ce 58 Cerium 140.1 Pr 59 Praseodymium 140.9 Nd 60 Neodymium 144.2 Pm 61 Promethium 145 Sm 62 Samarium 150.4 Eu 63 Europium 152.0 Gd 64 Gadolinium 157.3 Tb 65 Terbium 158.9 Dy 66 Dysprosium 162.5 Ho 67 Holmium 164.9 Er 68 Erbium 167.3 Tm 69 Thulium 168.9 Yb 70 Ytterbium 173.04 Lu 71 Lutetium 175.00																															
Ac 89 Actinium (227) Th 90 Thorium 232 Pa 91 Protactinium 231.0 U 92 Uranium 238.0 Np 93 Neptunium (237) Pu 94 Plutonium (244) Am 95 Americium (243) Cm 96 Curium (247) Bk 97 Berkelium (247) Cf 98 Californium (251) Es 99 Einsteinium (252) Fm 100 Fermium (257) Md 101 Mendelevium (258) No 102 Nobelium (259) Lr 103 Lawrencium (262)																															

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Specific heat and molar heat capacity (cont.)

Table 19.1 Molar Heat Capacities of Gases at Low Pressure

Type of Gas	Gas	C_V (J/mol · K)	C_p (J/mol · K)	$C_p - C_V$ (J/mol · K)	$\gamma = C_p/C_V$
Monatomic 	He	12.47	20.78	8.31	1.67
	Ar	12.47	20.78	8.31	1.67
Diatomic 	H ₂	20.42	28.74	8.32	1.41
	N ₂	20.76	29.07	8.31	1.40
	O ₂	20.85	29.17	8.31	1.40
	CO	20.85	29.16	8.31	1.40
Polyatomic 	CO ₂	28.46	36.94	8.48	1.30
	SO ₂	31.39	40.37	8.98	1.29
	H ₂ S	25.95	34.60	8.65	1.33

For a given substance, C_p and C_V are different. If the system can expand while heat is added, there is additional energy exchange through the performance of work by the system on its surroundings. If the volume is constant, the system does no work. So $C_p > C_V$.

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Problem. One mole of neon, a monatomic gas, starts out at conditions of standard temperature and pressure. The gas is heated at constant volume until its pressure is tripled, and subsequently further heated at constant pressure until it's volume is doubled. Assume that neon behaves as an ideal gas.

Compute the heat added to the gas for the entire process. (For neon, $C_V = \frac{3}{2}R$, and $C_p = \frac{5}{2}R$)

Answer:

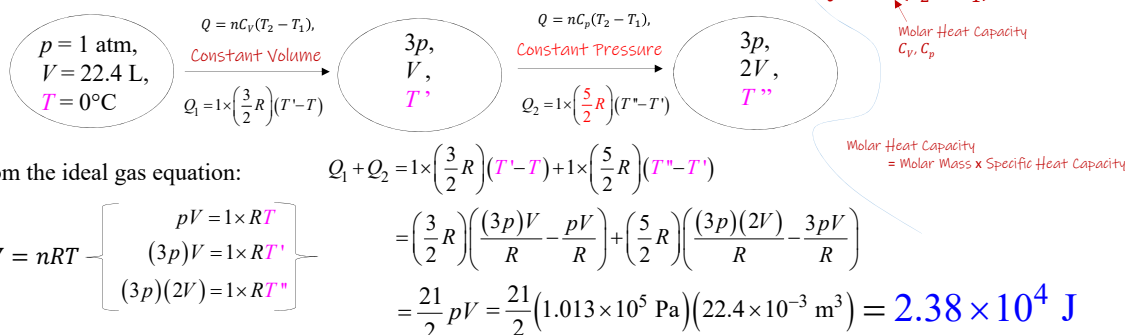
STP means at temperature of 0°C (273.15 K) and pressure 1 atm (1.013×10^5 N/m²).

At STP conditions, 1 mole of ideal gas has a volume of 22.4 litre.

(1 litre = 10 cm × 10 cm × 10 cm = 10⁻³ m³)

Let P , V , and T represent the initial values of pressure, volume, and temperature.

The flow of events is as follows:



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Table 17.3 Approximate Specific Heats and Molar Heat Capacities
(Constant Pressure)

Substance	Specific Heat, c (J/kg · K)	Molar Mass, M (kg/mol)	Molar Heat Capacity, C (J/mol · K)
Water (liquid)	4190	0.0180	75.4

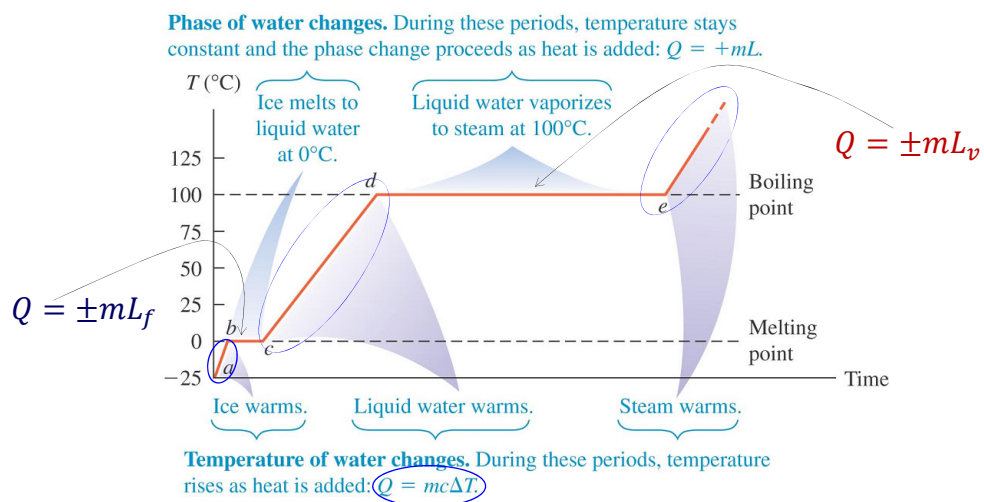
Specific Heat Capacity and Molar Heat Capacity are Measurements of Thermal Inertia



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Heat of fusion L_f and heat of vaporisation L_v (Latent Heat or Hidden Heat)

Heat is also involved in **phase changes**. For example,



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Heat of fusion L_f and heat of vaporisation L_v (cont.)**Table 17.4** Heats of Fusion and Vaporization

Substance	Normal Melting Point		Heat of Fusion, L_f (J/kg)	Normal Boiling Point		Heat of Vaporization, L_v (J/kg)
	K	°C		K	°C	
Helium	*	*	*	4.216	-268.93	20.9×10^3
Hydrogen	13.84	-259.31	58.6×10^3	20.26	-252.89	452×10^3
Nitrogen	63.18	-209.97	25.5×10^3	77.34	-195.8	201×10^3
Oxygen	54.36	-218.79	13.8×10^3	90.18	-183.0	213×10^3
Ethanol	159	-114	104.2×10^3	351	78	854×10^3
Mercury	234	-39	11.8×10^3	630	357	272×10^3
Water	273.15	0.00	334×10^3	373.15	100.00	2256×10^3
Sulfur	392	119	38.1×10^3	717.75	444.60	326×10^3
Lead	600.5	327.3	24.5×10^3	2023	1750	871×10^3
Antimony	903.65	630.50	165×10^3	1713	1440	561×10^3
Silver	1233.95	960.80	88.3×10^3	2466	2193	2336×10^3
Gold	1336.15	1063.00	64.5×10^3	2933	2660	1578×10^3
Copper	1356	1083	134×10^3	1460	1187	5069×10^3

*A pressure in excess of 25 atmospheres is required to make helium solidify. At 1 atmosphere pressure, helium remains a liquid down to absolute zero.

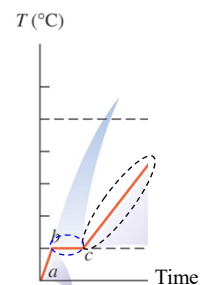
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Problem. Equal masses of two different liquids have the same temperature of 25°C. Liquid A has a freezing point of -68.0°C and a specific heat capacity of 1850 J/(kg °C). Liquid B has a freezing point of -96.0°C and a specific heat capacity of 2670 J/(kg °C). The same amount of heat must be removed from each liquid in order to freeze it into a solid at its respective freezing point. Determine the difference $L_{f,A} - L_{f,B}$ between the latent heat of fusion for these liquids.

Answer:

To freeze either liquid, heat must be removed to cool the liquid to its freezing point. In either case, the heat Q_1 that must be removed to lower the temperature of a substance of mass m by an amount ΔT is $Q_1 = m \times c \times \Delta T$, where c is the specific heat capacity. **Once the liquid has been cooled to its freezing point, additional heat must be removed to convert the liquid into a solid at the freezing point. The heat Q_2 that must be removed to freeze a mass m of liquid into a solid is $Q_2 = m \times L_f$, where L_f is the latent heat of fusion. The total heat required is $Q_1 + Q_2$.**

$$\begin{aligned}
 Mc_A (T_{i,A} - T_{f,A}) + ML_{f,A} &= Mc_B (T_{i,B} - T_{f,B}) + ML_{f,B} \\
 L_{f,A} - L_{f,B} &= c_B (T_{i,B} - T_{f,B}) - c_A (T_{i,A} - T_{f,A}) \\
 &= (2670 \text{ J/(kg} \cdot \text{°C)}) (25.0 \text{ °C} - (-96.0 \text{ °C})) \\
 &\quad - (1850 \text{ J/(kg} \cdot \text{°C)}) (25.0 \text{ °C} - (-68.0 \text{ °C})) \\
 &= 1.51 \times 10^5 \text{ J/kg}
 \end{aligned}$$



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Example.

A hot copper pot of mass 2.0 kg (including its copper lid) is at a temperature of 150°C. You pour 0.10 kg of cool water at 25°C into the pot, then quickly replace the lid so that no steam can escape. Find the final temperature of the pot and its contents, and determine the phase of the water (liquid, gas, or a **mixture**). Assume that no heat is lost to the surroundings.



Answer:

The specific heat of copper is $390 \text{ J kg}^{-1}\text{K}^{-1}$ and that of water is $4190 \text{ J kg}^{-1}\text{K}^{-1}$. So **suppose** the final temperature T is lower than 100°C and no water boils, then assuming no heat is lost to the surroundings, and heat loss = heat gain,

$$2 \times 390 \times (150 - T) = 0.1 \times 4190 \times (T - 25)$$

Heat lost by copper pot Heat gained by water

$$\rightarrow T \approx 106^\circ\text{C} \text{ (contradiction !!)}$$

WRONG!

Now suppose the **final temperature is 100°C** and x kg of water turns into steam at 100°C, then

$$2 \times 390 \times (150 - 100) = 0.1 \times 4190 \times (100 - 25) + x \times 2256 \times 10^3$$

Heat lost by copper pot Temperature of water elevated to 100°C x kg of 100°C water become 100°C steam

$$\rightarrow x \approx 3.4 \times 10^{-3} \text{ kg}$$

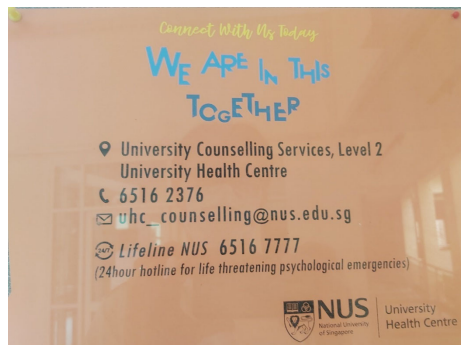
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Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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